

PHY: Physics Courses

Courses

PHY 1020 Conceptual Physics

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, Newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Meets General Education requirement in Natural Sciences.

PHY 2048 Calculus-Based Physics I

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: MAC 2311 AND MAC 2312*

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Meets General Education requirement in Natural Sciences.

PHY 2048L Calculus-Based Physics I Lab

College of Sci and Engineering, Department of Physics

1 sh (may not be repeated for credit)

Prerequisite: PHY 2048*

Calculus-Based Physics I Lab is intended to provide students with a fundamental understanding of the laws of physics with applications to our everyday life and major. This course also serves to develop cognitive and analytical skills necessary that are applicable to all majors. Selected experiments in mechanics, oscillatory motion, and heat. Equipment fees will be assessed.

PHY 2049 Calculus-Based Physics II

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: MAC 2312 AND PHY 2048

Continuation of PHY 2048. Electrostatics and magnetism; basic electric circuits; optics; selected topics in modern physics. Meets General Education requirement in Natural Sciences.

PHY 2049L Calculus-Based Physics II Lab

College of Sci and Engineering, Department of Physics

1 sh (may not be repeated for credit)

Prerequisite: PHY 2048/L AND PHY 2049*

Calculus-Based Physics II Lab is intended to provide students with a fundamental understanding of the laws of physics with applications to our everyday life and major. This course also serves to develop cognitive and analytical skills necessary that are applicable to all majors. Selected experiments in optics, electricity, and magnetism. Equipment fees will be assessed.

PHY 2053 Algebra-Based Physics I

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: (MAC 1105 AND MAC 1114) OR (MAC 1114 AND MAC 1140) OR MAC 1147 OR MAC 2311 OR 75 ALEKS Proctored test

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Meets General Education requirement in Natural Sciences.

PHY 2053L Algebra-Based Physics I Lab

College of Sci and Engineering, Department of Physics

1 sh (may not be repeated for credit)

Prerequisite: PHY 2053*

Algebra-Based Physics I Lab is intended to provide students with a fundamental understanding of the laws of physics with applications to our everyday life and major. This course also serves to develop cognitive and analytical skills necessary that are applicable to all majors. Selected experiments in mechanics, oscillatory motion, and heat. Equipment fees will be assessed.

PHY 2054 Algebra-Based Physics II

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 2053

Algebra-Based Physics II is the second of a two-semester sequence of physics topics chosen as an introduction to this science. This is an algebra and trigonometry based course. Light; optics; electricity and magnetism; circuits; elementary quantum theory; atomic, nuclear and particle physics. Meets General Education requirement in Natural Sciences.

PHY 2054L Algebra-Based Physics II Lab

College of Sci and Engineering, Department of Physics

1 sh (may not be repeated for credit)

Prerequisite: PHY 2054*

Algebra-Based Physics II Lab is intended to provide students with a fundamental understanding of the laws of physics with applications to our everyday life and major. This course also serves to develop cognitive and analytical skills necessary that are applicable to all majors. Selected experiments in optics, electricity, and magnetism. Equipment fees will be assessed.

PHY 2905 Directed Study

College of Sci and Engineering, Department of Physics

1-12 sh (may be repeated indefinitely for credit)

PHY 3106 Calculus-Based Physics III

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 2049

Laws of thermodynamics, wave phenomena, breakdown of classical physics, theory of relativity, quantization of charge, light, and energy, atomic structure.

PHY 3107 Calculus-Based Physics IV

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: MAP 2302 AND PHY 3106

Special topics in modern physics: quantum mechanics, atomic structure, molecular structure, atomic and molecular spectra, physics of solids, and band structure, nuclear structure, nuclear forces, radioactive decay and nuclear reactions, elementary particles, and fundamental interactions.

PHY 3220 Intermediate Mechanics

College of Sci and Engineering, Department of Physics

4 sh (may not be repeated for credit)

Prerequisite: MAP 2302* AND PHY 2048

Particle mechanics in 1, 2 and 3 dimensions for various forces. Central forces and celestial mechanics. Systems of many particles. Rigid body dynamics. Introduction to Lagrangian methods.

PHY 3424 Optics

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 2049

Geometrical, physical, and modern optics. Polarization, interference, diffraction, holography, and optical fibers. A grade of C- or better is required for all prerequisites.

PHY 3722C Electronics

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 2049

This is an introductory course in electronic design and circuitry with emphasis on common instrumentation. This course has both lecture and laboratory components.

PHY 3802L Intermediate Physics Lab

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 3106

Laboratory work in basic measurements of physical constants; experiments in electronics, modern physics, nuclear physics, optics, and solid state physics. Equipment fees will be assessed.

PHY 3905 Directed Study

College of Sci and Engineering, Department of Physics

1-12 sh (may be repeated indefinitely for credit)

PHY 4323 Electricity and Magnetism I

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: MAC 2313 AND MAP 2302 AND PHY 2049 AND PHZ 4113

Electrostatics, Gauss's Theorem, magnetic fields, Biot-Savart Law, electromagnetic induction, introduction to Maxwell's Equations, and electromagnetic waves. A grade of C- or better is required for prerequisite courses.

PHY 4325 Electricity and Magnetism II

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 4323

Continuation of PHY 4323 Electricity & Magnetism I. Maxwell's equations and electromagnetic waves in vacuum and in a medium, radiation from dipoles and antennas, transmission lines, wave guides, relativistic electrodynamics, Lienard-Weichert Potentials. A grade of C- or better in prerequisite courses is required.

PHY 4445 Lasers and Applications

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: MAC 2313 AND MAP 2302 AND PHY 2049

Introduction to lasers and applications covering topics on nature of light, photons, elements of semi-conductor physics, modulation of light, displays, laser principles, types of lasers and their design, photodetectors, fiber optics, optical communications. A grade of C- or better is required for all prerequisite courses.

PHY 4513 Thermal and Statistical Physics

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: MAC 2313 AND PHY 2048 AND PHZ 4113*

Laws of thermodynamics and their application to simple systems, kinetic theory of gases, introduction to the classical and quantum statistical mechanics of weakly interacting systems.

PHY 4604 Quantum Theory I

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 3107 AND PHZ 4113

This is the first semester of a two semester undergraduate level course covering the theory of quantum mechanics. This theory is the foundations of modern physics and is an introduction to the main concepts and tools for applying quantum mechanics to a variety of different problems.

PHY 4605 Quantum Theory II

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 4604

This is the second semester of a two semester undergraduate level course covering the theory of quantum mechanics. This theory is the foundations of modern physics. This course emphasizes the application of quantum mechanics to a variety of problems. Offered Spring semester only.

PHY 4822L Advanced Physics Lab

College of Sci and Engineering, Department of Physics

3 sh (may not be repeated for credit)

Prerequisite: PHY 3107 AND PHY 3802L

Advanced laboratory topics are treated. Modern physics laboratory equipment is used to introduce students to current laboratory practices.

PHY 4905 Directed Study

College of Sci and Engineering, Department of Physics

1-12 sh (may be repeated indefinitely for credit)

PHY 4910 Undergraduate Research

College of Sci and Engineering, Department of Physics

1-2 sh (may be repeated for up to 10 sh of credit)

Prerequisite: PHY 2049

Undergraduate experimental or theoretical research under the direction of physics faculty. Equipment fees will be assessed.

PHY 5905 Directed Study

College of Sci and Engineering, Department of Physics

1-12 sh (may be repeated indefinitely for credit)

* This course may be taken prior to or during the same term.